

PSimPy - Predictive and probabilistic simulation

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Simulate and Emulate

Robust GP emulator for your simulation

Create lightweight Surrogate model
and save precious compute time!

Case study ...

	training	evaluation	2500 evals
simulator	-	32 min	56 days
emulator	4.5 days	0.05 sec	2 min

Uncertainty Quantification

Quantify uncertainty. Identify key drivers.
Calibrate with data.

- Bayesian Calibration**
- Global Sensitivity Analysis** (Sobol indices)
- Efficient Sampling** (Saltelli, LHS, Metropolis Hastings)
- Active Learning**

**Want to build unified,
robust and sustainable
UQ workflows?**



Reference

[1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, “PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators,” in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.